

Week 7 Tutorial 1 – Converting IPv4 Addresses to Binary

Objectives

Part 1: Convert IPv4 Addresses from Dotted Decimal to Binary

Part 2: Use Bitwise ANDing Operation to Determine Network Addresses

Part 3: Apply Network Address Calculations

Background / Scenario – Make sure that you read this before moving on

IPv4 addresses consist of four numeric sections. The numbers are normally presented as decimal numbers separated by “dots”, this is known as “Dotted Decimal notation”. Underlying the dotted decimal scheme is the fact that each of the decimal numbers is actually an octet, an 8 bit binary string which sets the maximum and minimum values that can be shown in each section.

Every IPv4 address is comprised of two parts: a network portion and a host portion. The network portion of an address is the same for all devices that reside in the same network. The host portion identifies a specific individual host within a given network. The subnet mask is also a four section dotted decimal number which looks similar to an IP address; it is used to determine the network portion of an IP address.

Devices on the same network can communicate directly; devices on different networks require an intermediary Layer 3 device, such as a router or multilayer switch configured for routing, in order to communicate.

To understand the operation of devices on a network, we need to look at addresses the way devices do—in binary notation. To do this, we must convert the dotted decimal form of an IP address and its subnet mask to binary notation. After this has been done, we can use the bitwise AND operation to determine the network address.

This lab provides instructions on how to determine the network and host portion of IP addresses by converting addresses and subnet masks from dotted decimal to binary, and then using the bitwise ANDing operation. You will then apply this information to identify addresses in the network.

Part 1: Convert IPv4 Addresses from Dotted Decimal to Binary

In Part 1, you will convert decimal numbers to their binary equivalent. After you have mastered this activity, you will convert IPv4 addresses and subnet masks from dotted decimal to their binary form.

Step 1: Convert decimal numbers to their binary equivalent.

Fill in the missing details in Table 1 by converting the decimal number to an 8-bit binary number. The first number has been completed for you to confirm your method. Recall that the eight binary bit values in an octet are based on the powers of 2, and from left to right are 128, 64, 32, 16, 8, 4, 2, and 1.

(a method:

if the number is equal to or larger than 128, write down 1, subtract 128; if not write down 0

if the number is equal to or larger than 64, write down 1, subtract 64; if not write down 0

if the number is equal to or larger than 32, write down 1, subtract 32; if not write down 0

...

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Remember that the minimum value for an octet is 0, the maximum is 255. If you calculate numbers outside of the range 0 – 255, they are wrong!

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Of course, you could just use a calculator (or Internet lookup) to solve these problems but remember that you would only have the basic “four function” calculator provided for you in an exam so do think carefully how you would use it !

Answer 1: No, you cannot bring your own calculator to the exam,

Answer 2: No you will not have Internet access in the exam.

Table 1: convert decimal values to binary

Decimal	Binary
192	11000000
168	
10	
255	
2	

Step 2: Convert the IPv4 addresses to their binary equivalent.

An IPv4 address can be converted using the same technique you used above. Complete Table 2 with the binary equivalent of the addresses provided. To make your answers easier to read, separate the binary octets with a full stop.

Table 2: Show IP address in Binary notation

Decimal	Binary
192.168.10.10	11000000.10101000.00001010.00001010
209.165.200.229	
172.16.18.183	
10.86.252.17	
255.255.255.128	
255.255.192.0	

Part 2: Use Bitwise AND Operation to Determine Network Addresses

In Part 2, you will use the bitwise AND operation to calculate the network address for the provided host addresses.

You will first need to convert an IPv4 decimal address **and** subnet mask to their binary equivalent.

Once you have the binary form of the network address, convert it back to its (human readable) decimal form.

Note: The AND process compares the binary value in each bit position of the 32-bit host IP with the corresponding position in the 32-bit subnet mask. If there two 0s or a 0 and a 1, the AND result is 0. If there are two 1s, the result is a 1.

An example is shown in Table 3.

for information, the complete “Truth table” for logical AND operation:

0 AND 0 = 0
0 AND 1 = 0
1 AND 0 = 0
1 AND 1 = 1

Step 1: Determine the number of bits to use to calculate the network address.

Table 3: Example converting IP AND Subnet Mask to determine network address

Description	Decimal	Binary
IP Address	192.168.10.131 →	11000000.10101000.00001010.10000011
Subnet Mask	255.255.255.192 →	11111111.11111111.11111111.11000000
Network Address	192.168.10.128 ←	11000000.10101000.00001010.10000000

How do you determine what bits are used to calculate the network address?

In the example above, how many bits are used to calculate the network address?

Step 2: Use the AND operation to determine the network addresses:

- a. Enter the missing information into the table below:

Description	Decimal	Binary
IP Address	172.16.145.29	
Subnet Mask	255.255.0.0	
Network Address		

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- b. Enter the missing information into the table below:

Description	Decimal	Binary
IP Address	192.168.10.10	
Subnet Mask	255.255.255.0	
Network Address		

- c. Enter the missing information into the table below:

Description	Decimal	Binary
IP Address	192.168.68.210	
Subnet Mask	255.255.255.128	
Network Address		

- d. Enter the missing information into the table below:

Description	Decimal	Binary
IP Address	172.16.188.15	
Subnet Mask	255.255.240.0	
Network Address		

- e. Enter the missing information into the table below:

Description	Decimal	Binary
IP Address	10.172.2.8	
Subnet Mask	255.224.0.0	
Network Address		

Part 3: Apply Network Address Calculations

In Part 3, you must calculate the network address for the given IP addresses and subnet masks. After you have the network address, you should be able to determine the responses needed to complete the questions.

Step 1: Determine whether IP addresses are on same network.

- a. You are configuring two PCs for your network. PC-A is given an IP address of 192.168.1.18, and PC-B is given an IP address of 192.168.1.33. Both PCs are given a subnet mask of 255.255.255.240.

What is the network address for PC-A? _____

What is the network address for PC-B? _____

Will these PCs be able to communicate directly with each other? _____

What is the highest address that could be given to PC-B that allows it to be on the same network as PC-A? _____

- b. You are configuring two PCs for your network. PC-A is given an IP address of 10.0.0.16, and PC-B is given an IP address of 10.1.14.68. Both PCs have a subnet mask of 255.254.0.0.

What is the network address for PC-A? _____

What is the network address for PC-B? _____

Will these PCs be able to communicate directly with each other? _____

What is the lowest address that could be given to PC-B that allows it to be on the same network as PC-A?

Step 2: Identify the default gateway address.

- a. Your company has a policy to use the first IP address in a network as the default gateway address. A host on the local-area network (LAN) has an IP address of 172.16.140.24 and a subnet mask of 255.255.192.0.

What is the network address for this network?

What is the default gateway address for this host?

- b. Your company has a policy to use the first IP address in a network as the default gateway address. You have been instructed to configure a new server with an IP address of 192.168.184.227 and a subnet mask of 255.255.255.248.

What is the network address for this network?

What is the default gateway for this server?

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Reflection

Why is the subnet mask important in determining the network address?
